

CICD 作業報告

- 學號：314551180
 - 專案：cicd-lab (Fastify + TypeScript + Vitest)
 - Workflow 檔案：`.github/workflows/ci_314551180.yaml`
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一、CI Pipeline 說明

1.1 Workflow 主要內容

name: CI (314551180)

on: push

jobs:

ci:

name: Lint / Typecheck / Test

runs-on: ubuntu-latest

steps:

- name: Checkout
uses: actions/checkout@v4
- name: Setup Node.js
uses: actions/setup-node@v4
with:
node-version: '22'
cache: 'npm'
- name: Install dependencies
run: npm ci
- name: TypeScript typecheck
run: npm run typecheck
- name: Prettier check
run: npm run format:check
- name: Run tests (with JUnit report)
run: npx vitest run --reporter=default --reporter=junit --
outputFile=reports/junit.xml
- name: Publish test report
if: always()
uses: dorny/test-reporter@v1
with:
name: Vitest Tests
path: reports/junit.xml
reporter: jest-junit
fail-on-error: false
- name: Append summary to job page
if: always()
shell: bash
run: |
{
echo "### Vitest Result"

```

        if [ -f reports/junit.xml ]; then
            tests=$(grep -oE 'tests="[0-9]+' reports/junit.xml | head -1 |
grep -oE '[0-9]+')
            failures=$(grep -oE 'failures="[0-9]+' reports/junit.xml | head
-1 | grep -oE '[0-9]+')
            errors=$(grep -oE 'errors="[0-9]+' reports/junit.xml | head -1 |
grep -oE '[0-9]+')
            skipped=$(grep -oE 'skipped="[0-9]+' reports/junit.xml | head -1
| grep -oE '[0-9]+')
            echo "| Total | Failures | Errors | Skipped |"
            echo "|-----|-----|-----|-----|"
            echo "| ${tests:-0} | ${failures:-0} | ${errors:-0} |
${skipped:-0} |"
        else
            echo "_No JUnit report produced._"
        fi
    } >> "$GITHUB_STEP_SUMMARY"

- name: Upload JUnit report
  if: always()
  uses: actions/upload-artifact@v4
  with:
    name: junit-report
    path: reports/junit.xml
    if-no-files-found: ignore

```

1.2 Pipeline 設計說明

觸發條件： `on: push`，只要任何分支有 push 事件即自動執行，符合作業「push 時自動執行」要求。

執行環境： `ubuntu-latest` + Node.js 22（與 `package.json` 中 `engines.node` 範圍 `>=22 <25` 一致）。透過 `actions/setup-node@v4` 的 `cache: 'npm'` 加速重複執行的依賴安裝。

三項必要檢查（每個檢查獨立成 step，任一失敗 GitHub Actions 預設將整個 job 標為失敗）：

1. **TypeScript Typecheck** — `npm run typecheck`，等同 `tsc --noEmit`，純型別檢查不產生輸出檔。
2. **Prettier Check** — `npm run format:check`，使用 `prettier --check`，驗證所有檔案的格式。
3. **Test** — `npx vitest run` 同時帶 `default`（CLI 可讀輸出）與 `junit`（機器可讀報告）兩個 reporter，把結果寫到 `reports/junit.xml`。

測試結果展示策略：

機制	工具	顯示位置
測試報告 Tab	<code>dorny/test-reporter@v1</code> (GitHub Marketplace)	Actions run 結果頁的 "Vitest Tests" 分頁，可逐筆檢視通過/失敗 case
Job Summary	內建 <code>\$GITHUB_STEP_SUMMARY</code> + shell 解析 junit.xml	Actions run 主頁直接顯示 Markdown 統計表 (Total / Failures / Errors / Skipped)
Artifact	<code>actions/upload-artifact@v4</code>	提供 <code>junit-report</code> artifact 下載原始 XML

報告與 Summary 步驟都加上 `if: always()`，確保即便測試失敗仍會發佈報告，方便定位錯誤。

工具與策略總結：

- 使用專案既有的 npm scripts，避免在 workflow 重複定義指令。
- 採用 vitest 內建的 `junit reporter`（無需額外安裝套件）。
- 採用 GitHub Marketplace 上知名的 `dorny/test-reporter` 作為視覺化呈現。
- 用 `cache: 'npm'` 與 `npm ci` 確保依賴安裝可重現且快速。

二、CI 執行結果截圖

2.1 成功案例

hsuan1117 / cicc-lab

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CI (314551180)

chore: prettier format yaml/compose files #5

Re-run all jobs

Summary

All Jobs

Lint / Typecheck / Test

Vitest Tests

Run details

Usage

Workflow file

Triggered via push 1 minute ago

hsuan1117 pushed 8fc32af main

Status

Success

Total duration

26s

Artifacts

1

ci_314551180.yaml

on: push

Lint / Typecheck / Test 17s

Lint / Typecheck / Test summary

Vitest Result

Total	Failures	Errors	Skipped
2	0	0	0

Job summary generated at run-time

hsuan1117 / cicc-lab

Q Type to search

<> Code Pull requests Agents Actions Projects Wiki Security and quality Insights Settings

CI (314551180)

chore: prettier format yaml/compose files #5

Re-run all jobs

Summary

All Jobs

Lint / Typecheck / Test

Vitest Tests

Run details

Usage

Workflow file

GitHub Actions / Vitest Tests

succeeded 3 minutes ago in 0s

2 passed, 0 failed and 0 skipped

tests 2 passed

reports/junit.xml

2 tests were completed in 113ms with 2 passed, 0 failed and 0 skipped.

Test suite	Passed	Failed	Skipped	Time
test/app.test.ts	2			113ms

test/app.test.ts

test/app.test.ts

Fastify app > GET /health returns ok status

Fastify app > GET / returns app message and version

View more details on GitHub Actions

三、失敗案例說明

為了驗證「任一檢查失敗時 pipeline 顯示失敗」，故意在 `test/app.test.ts` 中製造一個測試失敗：

3.1 錯誤製造方式（Test 失敗）

將某個測試的預期值改錯，例如：

```
// test/app.test.ts
it('GET /health returns ok', async () => {
  const res = await app.inject({ method: 'GET', url: '/health' });
  expect(res.statusCode).toBe(500); // 故意改錯：實際應為 200
});
```

3.2 Pipeline 失敗截圖

← CI (314551180)

fix(ci): use java-junit parser for vitest output #7

Re-run jobs

Summary

All jobs

Lint / Typecheck / Test

Vitest Tests

Run details

Usage

Workflow file

GitHub Actions / Vitest Tests

succeeded now in 0s

1 passed, 1 failed and 0 skipped

tests 1 passed, 1 failed

reports/junit.xml

2 tests were completed in 116ms with 1 passed, 1 failed and 0 skipped.

Test suite	Passed	Failed	Skipped	Time
test/app.test.ts	1	1		116ms

test/app.test.ts

Fastify app > GET /health returns ok status
AssertionError: expected 200 to be 500 // Object.is equality
Fastify app > GET / returns app message and version

ANNOTATIONS

Check failure on line 0 in reports/junit.xml
github-actions / Vitest Tests
test/app.test.ts ▶ Fastify app > GET /health returns ok status
Failed test found in:
reports/junit.xml
Error:
AssertionError: expected 200 to be 500 // Object.is equality

Raw output

← CI (314551180)

fix(ci): use java-junit parser for vitest output #7

Re-run jobs

Summary

All jobs

Lint / Typecheck / Test

Vitest Tests

Run details

Usage

Workflow file

Triggered via push 1 minute ago

hsuan1117 pushed → f466cc6 main

Status

Failure

Total duration

19s

Artifacts

1

ci_314551180.yaml

on: push

Lint / Typecheck / Test 15s

Lint / Typecheck / Test summary

Vitest Result

Total	Failures	Errors	Skipped
2	1	0	0

Job summary generated at run-time

Annotations

2 errors and 1 warning

- ✖ test/app.test.ts ▶ Fastify app > GET /health returns ok status: reports/junit.xml#L0
Failed test found in: reports/junit.xml Error: AssertionError: expected 200 to be 500 // Object.is equality
- ✖ Lint / Typecheck / Test

3.3 錯誤原因與修正方式

- **原因：** `/health` route 實作回傳 HTTP 200，但測試 `expect` 為 500，導致 vitest 斷言失敗，`Run tests` step exit code 非 0，GitHub Actions 將整個 job 標記為 failure。
- **修正：** 把 `expect(res.statusCode).toBe(500)` 改回 `expect(res.statusCode).toBe(200)`，重新 push，pipeline 再次跑過即恢復綠燈。

同樣的失敗策略也適用於：

- **TypeScript 型別錯誤：** 在 `src/server.ts` 加 `const x: number = 'str';`，`npm run typecheck` 會在 ts(2322) 失敗。
- **Prettier 格式錯誤：** 在任一 `.ts` 檔加多餘空白後不執行 `npm run format`，`format:check` 會列出 code style issues 並 exit 1。

四、結論

本次作業透過 GitHub Actions 建立了一條符合 push 觸發、三項自動檢查（typecheck / prettier / test）、結果可視化（dorny test-reporter + Job Summary + artifact）的 CI Pipeline。任一檢查失敗皆會即時反映在 Actions 結果頁，符合作業全部要求。